

Srikanth Prabala

Senior Applied Scientist, Amazon Ads

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RESEARCH INTERESTS

Post-training of large language models: supervised fine-tuning, preference optimization, and policy-gradient reinforcement learning. Agentic systems: multi-turn tool use, reward specification for long-horizon tasks, and harness design. Evaluation methodology and measurement reliability for large models. Speech and spoken language processing. Inference and serving efficiency for large language models.

PROFESSIONAL EXPERIENCE

Senior Applied Scientist

May 2025 – Present

Amazon Ads

Seattle, WA

- Post-train the large language models behind Amazon Ads Agent, Amazon's agentic advertising assistant, which executes multi-step workflows over advertiser campaigns with tool use and human approval
- Define the post-training methodology for the capability: model selection, supervised fine-tuning, preference optimization (DPO), and policy-gradient reinforcement learning (GRPO), together with the evaluation standards used to accept or reject a checkpoint
- Build end-to-end post-training pipelines spanning synthetic data generation, distributed training on verl and NeMo-RL, and evaluation via LM Eval Harness and LLM-as-a-judge, at the scale of 100B+ parameter mixture-of-experts models; the resulting tooling and training configurations are used by other teams
- Train tool-using agents with reinforcement learning, with emphasis on reward specification for multi-turn trajectories, trace-level diagnosis across successive training rounds, and retention of general capabilities such as instruction following and tool use
- Establish evaluation protocols separating model-intrinsic variance from variance introduced by the inference configuration, so that reported differences can be judged against a known measurement floor

Senior Software Engineer

Aug 2024 – Apr 2025

Amazon Web Services, Amazon Bedrock

Seattle, WA

- Led the technical roadmap and integration architecture for bringing Anthropic's Claude models onto Amazon Bedrock, coordinating across model hosting, runtime, and frontend service teams
- Delivered document and image understanding support for Claude models on Bedrock, including the Citations API and PDF processing capability, with an in-memory-only architecture cleared through application security review and penetration testing
- Reduced large language model inference latency through end-to-end profiling of the serving stack, isolating bottlenecks across prefill, content safety classification, and decoding
- Extended the serving architecture to use capacity in AWS Local Zones alongside Availability Zones, relieving accelerator capacity constraints in high-demand regions

Senior Software Engineer

Mar 2019 – Aug 2024

Amazon Web Services, Contact Lens and Amazon Transcribe

Seattle, WA

- Founding engineer on a machine learning service for contact center analytics, built on speech transcription and natural language processing over conversational transcripts
- Led an engineering team delivering real-time streaming call analytics, launched across two separate AWS re:Invent releases
- Designed and productionized natural language processing models for classification, semantic phrase matching, summarization, sentiment analysis, and theme detection over large volumes of conversational data
- Architected low-latency streaming inference systems, addressing session lifecycle management, fault tolerance, and long-running bidirectional streams, scaled to enterprise deployments with thousands of concurrent sessions

Software Engineering Intern

May 2018 – Aug 2018

Google

Mountain View, CA

- Built a machine learning model to reduce compute consumption in large-scale data processing pipelines

Software Engineer, Search and Recommendation

Jul 2016 – Jul 2017

TiVo

Bangalore, India

- Built knowledge graph and natural language processing systems for search and recommendation, including entity extraction and semantic matching

Research Intern

May 2015 – Jul 2015

Samsung R&D Institute India

Bangalore, India

- Research on automatic speech recognition and on-device machine learning

ISSUED PATENTS

Both patents are classified across **machine learning, natural language processing, information retrieval, and speech and spoken language processing.**

- **Programmatic Theme Detection in Contacts Analytics Service**

US Patent No. 12,062,368 B1. Issued 13 August 2024. Assignee: Amazon Technologies, Inc.

Machine learning system for detecting, aggregating, and ranking recurring themes across large volumes of conversational transcripts.

Classifications: G06N 20/00 (machine learning); G10L 15/1822 (natural language modelling in recognition); G10L 15/063 (model training); G10L 15/26 (speech-to-text); G06Q 30/01 (customer relationship data processing).

- **Systems and Methods to Analyze Customer Contacts**

US Patent No. 11,862,148 B2. Issued 2 January 2024. Assignee: Amazon Technologies, Inc.

Cloud service for analyzing multimodal conversational data using transcription and natural language processing to produce structured, queryable insights.

Classifications: G06N 5/04 (inference and knowledge-based systems); G06F 16/61 (information retrieval and indexing); G10L 15/1815 (natural language modelling in recognition); G10L 15/22 (dialogue procedures); G10L 15/30 (distributed recognition); G10L 15/32 (multiple recognisers).

Searched art additionally spans G06F 40/20–40/289 (NLP, parsing, entity recognition), G06F 16/3329 (natural language query), G10L 25/63 (affective analysis).

Inventorship corrected post-grant under 35 U.S.C. §256; Certificate of Correction issued 1 October 2024.

PROFESSIONAL SERVICE

- *Peer review and program committee service in progress; entries added as roles are confirmed.*

TECHNICAL EXPERTISE

Post-training and alignment: supervised fine-tuning, preference optimization (DPO), policy-gradient reinforcement learning (GRPO), reward design, instruction tuning, parameter-efficient fine-tuning

Agentic systems and evaluation: multi-turn tool use, harness and scaffolding design, trajectory analysis, reward specification for long-horizon tasks, benchmark design, measurement reliability, LLM-as-a-judge

Speech and language: automatic speech recognition, speech-to-text pipelines, spoken language understanding, entity and key-phrase extraction, sentiment analysis, topic and theme detection

Inference and systems: distributed training, mixture-of-experts architectures, speculative decoding, KV cache behaviour, tensor parallelism, low-latency streaming inference

Frameworks: PyTorch, verl, NeMo-RL, vLLM, SGLang, Ray, Megatron, FSDP

EDUCATION

University of Massachusetts Amherst

Amherst, MA

Master of Science in Computer Science

Sep 2017 – Dec 2018

Indian Institute of Technology Madras

Chennai, India

Bachelor of Technology in Electrical Engineering, Minor in Management Studies

Aug 2012 – May 2016

HONORS

- Kishore Vaigyanik Protsahan Yojana (KVPY) Scholar, Government of India